Personally implemented formulae in WannierBerri
Working handbook: geometry, covariant derivatives, and interpolation

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This working copy consolidates the results in `formulae.tex`, `Wannier.tex`, and `QGD.tex`. It establishes common conventions for the subsequent reconciliation of the v1.7 and v1.8 calculators. Formula identities, representation assumptions, and implementation status are stated separately. The original source notes remain unchanged. Source corrections and unfinished results are recorded in Appendix A.

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I. BASIC INGREDIENTS

A. Notation, units, and scope

We use Cartesian wave-vector derivatives $\partial_a = \partial/\partial k_a$ and $\epsilon_{xyz} = +1$. Repeated Cartesian indices are summed only in explicitly contracted expressions; free indices remain free. The index convention throughout this handbook is

$$\alpha, \beta, \gamma \in \{A, B\}, \qquad m, n \in A, \qquad l, p \in B, \qquad j, k \in \text{Full} = A \oplus B. \tag{1}$$

Greek indices label blocks; A, B themselves denote the chosen blocks. Latin m, n, l, p, j, k label states, whereas $a, b, c, d, e, \ldots$ label Cartesian directions. Band and block indices are subscripts; Cartesian components and derivative indices of matrix elements are superscripts, as in $X^{ab:d}_{mn}$ and $\Delta^{,d}_{jk}$. Differential operators retain their usual subscripts, e.g. ∂_a and ∇_a. Primes on band indices do not change their index space. In the matrix-element formulae, every primed band index is implicitly summed once over its assigned space: $m', n', m'', n'' \in A$, $l', p', l'', p'' \in B$, and $j', k', j'', k'' \in \text{Full}$. Thus primed dummy indices do not require an explicit summation sign. Free indices m, n are not summed unless a sum or trace is written explicitly. For example,

$$(X_{AB}Y_{BA})_{mn} = \sum_{l \in B} X_{ml}Y_{ln} \equiv X_{ml'}Y_{l'n}, \qquad (X_{AA}Y_{AA})_{mn} = X_{mn'}Y_{n'n}. \tag{2}$$

The first product sums over B; the second sums over A. For full-space products and commutators, the same convention gives

$$(XY)_{jk} = X_{jj'}Y_{j'k}, \qquad [X, Y]_{jk} = X_{jj'}Y_{j'k} - Y_{jj'}X_{j'k}. \tag{3}$$

Explicit matrix elements accompany the compact block identities below. The blocks are separated by a nonzero gap at the wave vector under consideration; degeneracies within a block are allowed. An isolated-band formula selects $A = \{n\}$, with its complement in B; unrestricted dummy sums use j, k, with exclusions written explicitly. Here Full means the full band space of the Hamiltonian being considered. For a finite Wannier model it is the represented space, not all physical states; external contributions are treated separately below. In Wannier-gauge matrices, j, k label basis functions spanning that same represented space, rather than energy eigenstates. The spatial dimension is denoted by d_{BZ}, independently of Cartesian index d.

$$\Delta_{jk} = \varepsilon_j - \varepsilon_k, \qquad \int [\mathrm{d}\boldsymbol{k}] = \int_{\text{BZ}} \frac{\mathrm{d}^{d_{\text{BZ}}}k}{(2\pi)^{d_{\text{BZ}}}}, \qquad e > 0, \quad q_{\text{electron}} = -e. \tag{4}$$

The geometric expressions use energies and wave vectors, without silently setting $\hbar = 1$. In particular,

$$v^a_n = \frac{1}{\hbar}\partial_a \varepsilon_n, \qquad E^a_n = \partial_a \varepsilon_n. \tag{5}$$

If energy is in eV and length in Å, E^a_n is in eV Å; it is not yet a velocity in m/s. The conversion is $v^a_n = (1\,\text{eV in J})(1\,\text{Å in m})E^a_n/\hbar$.

Symbol	Meaning	Relation to the source notes
Q^{ab}_n	quantum geometric tensor (QGT)	QG; called QMT in `formulae.tex`
G^{ab}_n	single-band quantum metric	uppercase throughout this handbook
$\widetilde{G}^{ab}_{mn}$	subspace quantum metric	matrix elements in the active space
$\mathcal{G}^{ab}_n$	Berry connection polarizability (BCP)	original normalization retained
D^a	$U^\dagger \partial_a U$	coefficient-space connection
$\bar{A}^a$	$U^\dagger A^{W,a} U$	rotated Wannier connection
M^a	$\bar{A}^a + \mathrm{i}D^a$	auxiliary connection in BCP formulae
$\widetilde{\mathcal{E}}^{ab}$	energy-weighted geometric matrix	renamed from $\widetilde{G}$ in `Wannier.tex`
$\mathfrak{m}^a$	interband magnetic kernel	separates magnetic M from BCP M

B. Berry connection

Single band Berry connection (Abelian Berry connection):

$$A^a_n(\boldsymbol{k}) = i\langle u_n(\boldsymbol{k})|\partial_{k_a} u_n(\boldsymbol{k})\rangle \equiv i\langle u_n|\partial_a u_n\rangle \tag{6}$$

Matrix Berry connection (with interband elements for $j \neq k$):

$$A^a_{jk}(\boldsymbol{k}) = i\langle u_j(\boldsymbol{k})|\partial_{k_a} u_k(\boldsymbol{k})\rangle \equiv i\langle u_j|\partial_a u_k\rangle \tag{7}$$

The matrix Berry connection is Hermitian, $(A^a)^\dagger = A^a$. Its full matrix transforms inhomogeneously under a wave-vector-dependent change of basis. Off-block elements transform covariantly under independent rotations within A and B.

C. Quantum geometric tensor (QGT)

Quantum geometry tensor:

$$\begin{aligned} Q^{ab}_n(\boldsymbol{k}) &= \langle \partial_{k_a} u_n(\boldsymbol{k})|(1 - |u_n(\boldsymbol{k})\rangle\langle u_n(\boldsymbol{k})|)|\partial_{k_b} u_n(\boldsymbol{k})\rangle \\ &\equiv \langle \partial_a u_n|(1 - |u_n\rangle\langle u_n|)|\partial_b u_n\rangle \\ &= \sum_{j(\neq n)} \langle \partial_a u_n|u_j\rangle\langle u_j|\partial_b u_n\rangle \\ &= \sum_{j(\neq n)} A^a_{nj} A^b_{jn} \\ &= G^{ab}_n(\boldsymbol{k}) - \frac{i}{2}\Omega^{ab}_n(\boldsymbol{k}) \end{aligned} \tag{8}$$

Quantum metric tensor:

$$G^{ab}_n(\boldsymbol{k}) = \mathrm{Re}\, Q^{ab}_n(\boldsymbol{k}) = \mathrm{Re} \sum_{j(\neq n)} A^a_{nj} A^b_{jn} \tag{9}$$

Berry curvature tensor:

$$\Omega^{ab}_n(\boldsymbol{k}) = -2\,\mathrm{Im}\, Q^{ab}_n(\boldsymbol{k}) = -2\,\mathrm{Im} \sum_{j(\neq n)} A^a_{nj} A^b_{jn} \tag{10}$$

Berry curvature vector:

$$\Omega^a_n(\boldsymbol{k}) = \frac{1}{2}\epsilon^{abc}\Omega^{bc}_n(\boldsymbol{k}) = -\epsilon^{abc}\,\mathrm{Im} \sum_{j(\neq n)} A^b_{nj} A^c_{jn} = \epsilon^{abc}\partial_b A^c_n \tag{11}$$

The band sum in Eq. (8) requires a complete complement to n. A finite-model band sum gives the geometry of that model; missing states in a Wannier representation require the external terms of Sec. III. The metric is symmetric and positive semidefinite, whereas $\Omega^{ab}_n = -\Omega^{ba}_n$.

D. Berry connection polarizability (BCP)

Berry connection polarizability [2]:

$$\mathcal{G}^{ab}_n(\boldsymbol{k}) = \frac{1}{2} \sum_{j(\neq n)} \frac{A^a_{nj} A^b_{jn} + A^b_{nj} A^a_{jn}}{\varepsilon_n - \varepsilon_j} = \mathrm{Re} \sum_{j(\neq n)} \frac{A^a_{nj} A^b_{jn}}{\varepsilon_n - \varepsilon_j} \tag{12}$$

The normalization in Eq. (12) is retained from the original notes. Any response formula imported from another convention must first be converted to this definition; a factor of two must not be inferred from the name BCP alone. The units of Q, G, and Ω are length squared; those of $\mathcal{G}$ are length squared divided by energy.

II. COVARIANT DERIVATIVES AND WANNIER CONVENTIONS

A. Gauge dictionary and scope

Here a gauge specifies a wave-vector-dependent orthonormal frame for a fixed space of electronic states. It concerns Bloch-state and basis choices, rather than the electromagnetic gauge of an applied field. The relevant choices are summarized below; their transformation laws are specified after the Wannier and Hamiltonian conventions.

Choice	Defining frame or transformation	Meaning in this handbook
Wannier gauge	$\{\lvert u_j^W\rangle\}$ from a chosen set of Wannier functions	Smooth interpolation basis; $\mathcal{H}^W$ is generally not diagonal.
Hamiltonian gauge	$\lvert u_n\rangle = \sum_j \lvert u_j^W\rangle U_{jn}$	Energy eigenbasis; $U^\dagger \mathcal{H}^W U = \varepsilon$.
Block gauge	$W = \mathrm{diag}(W_A, W_B)$	Independent rotations within fixed active and complementary subspaces; the Hamiltonian remains block diagonal.
Single-band phase	$\lvert u_n\rangle \mapsto e^{\mathrm{i}\phi_n}\lvert u_n\rangle$	Residual $U(1)$ freedom of a nondegenerate energy eigenstate.
Parallel transport	A selected connection vanishes along a chosen path	A local gauge-fixing condition; the connection and transport direction must be specified.
Wannier-center convention	Inclusion or omission of $e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{t}_j}$	A particular rephasing of the Wannier basis, requiring consistent changes to all matrix data.

The Hamiltonian and Wannier gauges describe the same represented space. Changing the disentanglement window or the represented subspace changes the model or projector, and is not merely a gauge transformation. No maximally localized gauge is required by the algebra below; localization is a construction choice for the Wannier functions.

B. Wannier and Hamiltonian gauges

With Wannier centers $\boldsymbol{t}_j$, define

$$|u_{k\boldsymbol{k}}^W\rangle = \sum_{\boldsymbol{R}} e^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{t}_k-\boldsymbol{r})}|\boldsymbol{R}k\rangle, \tag{13}$$

$$\mathcal{H}_{jk}^W(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_{jk}}\langle \boldsymbol{0}j|\hat{\mathcal{H}}|\boldsymbol{R}k\rangle, \qquad \boldsymbol{R}_{jk} = \boldsymbol{R}+\boldsymbol{t}_k-\boldsymbol{t}_j, \tag{14}$$

$$\mathcal{H}^W U = U\varepsilon, \qquad |u_n\rangle = \sum_k |u_k^W\rangle U_{kn}, \qquad \bar{X} = U^\dagger X^W U. \tag{15}$$

Normalization of the Bloch sums is understood. Fourier derivatives follow by multiplying every real-space term by $\mathrm{i}R_{jk}^a$, with one factor for each derivative.

For the coefficient vectors (columns of U),

$$D^a = U^\dagger \partial_a U, \qquad (D^a)^\dagger = -D^a, \qquad (D_{jk}^a)^* = -D_{kj}^a, \tag{16}$$

$$D_{ln}^a = \frac{\mathcal{H}_{ln}^{;a}}{\varepsilon_n - \varepsilon_l}, \quad l \in B,\ n \in A, \qquad \mathcal{H}^{;a} = U^\dagger(\partial_a \mathcal{H}^W)U, \tag{17}$$

$$|\partial_a u_n\rangle = \sum_k |\partial_a u_k^W\rangle U_{kn} + \sum_j |u_j\rangle D_{jn}^a, \tag{18}$$

$$A^{H,a} = \bar{A}^a + \mathrm{i}D^a \equiv M^a, \qquad A_{jk}^{W,a} = \mathrm{i}\langle u_j^W|\partial_a u_k^W\rangle. \tag{19}$$

The connection and rotation in matrix elements are

$$\begin{aligned} D_{jk}^a &= (U^\dagger)_{jj'}\partial_a U_{j'k}, \\ \bar{X}_{jk} &= (U^\dagger)_{jj'}X_{j'k'}^W U_{k'k}, \qquad M_{jk}^a = \bar{A}_{jk}^a + \mathrm{i}D_{jk}^a. \end{aligned} \tag{20}$$

Diagonal entries of D depend on the phase choice. Equation (17) is used only across separated blocks, never to divide by a zero gap within a degenerate group. In general, $D^a_{jk} \neq \langle u_j|\partial_a u_k\rangle$: the latter also contains the derivative of the Wannier basis.

C. Gauge transformations and invariant quantities

a. Rotations of the band frame. Hold the Wannier basis fixed and let $U' = UW$, where $W = \mathrm{diag}(W_A, W_B)$ is unitary. Matrix multiplication below is over band indices; all matrices and rotations can depend on $\boldsymbol{k}$. For an operator matrix X and the connections,

$$X' = W^\dagger X W, \qquad D'^a = W^\dagger D^a W + W^\dagger \partial_a W, \tag{21}$$

$$\bar{A}'^a = W^\dagger \bar{A}^a W, \qquad A'^{H,a} = W^\dagger A^{H,a} W + \mathrm{i} W^\dagger \partial_a W. \tag{22}$$

In full-space matrix elements these laws read

$$\begin{aligned} X'_{jk} &= (W^\dagger)_{jj'} X_{j'k'} W_{k'k}, \\ D'^a_{jk} &= (W^\dagger)_{jj'} D^a_{j'k'} W_{k'k} + (W^\dagger)_{jj'} \partial_a W_{j'k}, \\ \bar{A}'^a_{jk} &= (W^\dagger)_{jj'} \bar{A}^a_{j'k'} W_{k'k}, \\ A'^{H,a}_{jk} &= (W^\dagger)_{jj'} A^{H,a}_{j'k'} W_{k'k} + \mathrm{i}(W^\dagger)_{jj'} \partial_a W_{j'k}. \end{aligned} \tag{23}$$

Thus the full A^H and D are connections, not covariant operator matrices. Their off-block parts are covariant because the inhomogeneous term is block diagonal:

$$X'_{AB} = W_A^\dagger X_{AB} W_B, \quad D'^a_{AB} = W_A^\dagger D^a_{AB} W_B, \quad M'^a_{AB} = W_A^\dagger M^a_{AB} W_B. \tag{24}$$

With $W_{nl} = W_{ln} = 0$, the same expressions reduce to

$$\begin{aligned} X'_{ml} &= (W^\dagger)_{mm'} X_{m'l'} W_{l'l}, \\ D'^a_{ml} &= (W^\dagger)_{mm'} D^a_{m'l'} W_{l'l}, \qquad M'^a_{ml} = (W^\dagger)_{mm'} M^a_{m'l'} W_{l'l}, \\ X'_{mn} &= (W^\dagger)_{mm'} X_{m'n'} W_{n'n}, \qquad \mathcal{H}'_{mn} = (W^\dagger)_{mm'} \varepsilon_{m'} W_{m'n}. \end{aligned} \tag{25}$$

This is the gauge freedom used by the block-covariant construction of Ref. [1]. An arbitrary W_A may mix different energies within the active group. Then $\mathcal{H}'_{AA} = W_A^\dagger \varepsilon_A W_A$ need not be diagonal, although the physical subspace and its spectrum are unchanged. The rotated frame is an admissible block gauge, but is generally no longer the Hamiltonian gauge.

To stay in the Hamiltonian gauge, a continuous rotation must preserve the energy eigenspaces. Nondegenerate eigenstates retain independent phases; an exactly degenerate eigenspace permits a full unitary rotation. A numerical grouping of nearly degenerate bands does not make their distinct energies exactly equal. Explicit denominators $1/(\varepsilon_j - \varepsilon_k)$ refer to the energy eigenbasis. In a general block frame their covariant meaning is the inverse Sylvester operator of Eq. (86).

b. Single-band phase freedom. For $W_{jk} = \delta_{jk} e^{\mathrm{i}\phi_j}$, Eqs. (21) and (22) imply

$$A'^a_j = A^a_j - \partial_a \phi_j, \qquad D'^a_{jj} = D^a_{jj} + \mathrm{i}\partial_a \phi_j, \tag{26}$$

$$A'^a_{jk} = e^{\mathrm{i}(\phi_k - \phi_j)} A^a_{jk}, \qquad j \neq k. \tag{27}$$

Therefore $A^a_{nj} A^b_{jn}$, Q^{ab}_n, G^{ab}_n, Ω^{ab}_n, and $\mathcal{G}^{ab}_n$ are phase invariant for isolated bands. Their individual band labels do not define invariant quantities under arbitrary mixing of a degenerate multiplet; use the corresponding subspace objects there.

c. Changing the Wannier basis while holding Bloch eigenstates fixed. Write the Wannier frame as a row of kets, so that $|u^W\rangle \mathsf{S}$ denotes a linear combination of its columns. For a smooth unitary $\mathsf{S}(\boldsymbol{k})$ within the same represented space,

$$\begin{aligned} |u^{W'}\rangle &= |u^W\rangle \mathsf{S}, \qquad U' = \mathsf{S}^\dagger U, \qquad |u^{H'}\rangle = |u^H\rangle, \\ \mathcal{H}^{W'} &= \mathsf{S}^\dagger \mathcal{H}^W \mathsf{S}, \qquad A^{W',a} = \mathsf{S}^\dagger A^{W,a} \mathsf{S} + \mathrm{i}\mathsf{S}^\dagger \partial_a \mathsf{S}. \end{aligned} \tag{28}$$

Equivalently, for each basis state and matrix element,

$$\begin{aligned} |u_j^{W'}\rangle &= |u_{j'}^{W}\rangle \mathsf{S}_{j'j}, \qquad U'_{jn} = (\mathsf{S}^\dagger)_{jj'} U_{j'n}, \\ \mathcal{H}^{W'}_{jk} &= (\mathsf{S}^\dagger)_{jj'} \mathcal{H}^{W}_{j'k'} \mathsf{S}_{k'k}, \\ A^{W',a}_{jk} &= (\mathsf{S}^\dagger)_{jj'} A^{W,a}_{j'k'} \mathsf{S}_{k'k} + \mathrm{i}(\mathsf{S}^\dagger)_{jj'} \partial_a \mathsf{S}_{j'k}. \end{aligned} \tag{29}$$

Define $\mathcal{K}^a = U^\dagger(\partial_a \mathsf{S})\mathsf{S}^\dagger U$. In the unchanged physical band frame,

$$D'^a = D^a - \mathcal{K}^a, \qquad \bar{A}'^a = \bar{A}^a + \mathrm{i}\mathcal{K}^a, \qquad \bar{A}'^a + \mathrm{i}D'^a = A^{H,a}. \tag{30}$$

The compensating terms can also be read element by element:

$$\begin{aligned} \mathcal{K}^a_{jk} &= (U^\dagger)_{jj'}(\partial_a \mathsf{S}_{j'k'})(\mathsf{S}^\dagger)_{k'j''} U_{j''k}, \\ D'^a_{jk} &= D^a_{jk} - \mathcal{K}^a_{jk}, \qquad \bar{A}'^a_{jk} = \bar{A}^a_{jk} + \mathrm{i}\mathcal{K}^a_{jk}, \qquad \bar{A}'^a_{jk} + \mathrm{i}D'^a_{jk} = A^{H,a}_{jk}. \end{aligned} \tag{31}$$

The split into internal and external terms can consequently depend on the Wannier representation. A complete expression for the same physical states remains unchanged when every basis-derivative term is transformed consistently. Dropping external terms is a model approximation, not a general gauge fixing.

d. Wannier-center phase convention. Let $|u_j^{W,0}\rangle$ omit the center phase and define $\mathsf{S}_t = \mathrm{diag}(e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{t}_j})$. The convention in Eq. (14) satisfies

$$\begin{aligned} |u_j^{W,t}\rangle &= e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{t}_j}|u_j^{W,0}\rangle, & \mathcal{H}^{W,t} &= \mathsf{S}_t^\dagger \mathcal{H}^{W,0}\mathsf{S}_t, \\ A^{W,t,a} &= \mathsf{S}_t^\dagger A^{W,0,a}\mathsf{S}_t - \mathsf{t}^a, & \mathsf{t}^a_{jk} &= t^a_j \delta_{jk}. \end{aligned} \tag{32}$$

In particular, the center convention gives

$$\begin{aligned} \mathcal{H}^{W,t}_{jk} &= e^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{t}_k - \boldsymbol{t}_j)}\mathcal{H}^{W,0}_{jk}, \\ A^{W,t,a}_{jk} &= e^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{t}_k - \boldsymbol{t}_j)}A^{W,0,a}_{jk} - t^a_j\delta_{jk}. \end{aligned} \tag{33}$$

The Hamiltonian Fourier phase changes from $e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}}$ to $e^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{R}+\boldsymbol{t}_k-\boldsymbol{t}_j)}$. This is a special case of Eq. (28), including the connection shift. Changing only the hopping phase while keeping all position and derivative matrices unchanged is inconsistent. The resulting periodicity across reciprocal-lattice boundaries can differ between frames; physical quantities are compared using the corresponding boundary transformations.

e. Covariance, invariance, and projectors. A covariant matrix changes its components as $X'_{AA} = W_A^\dagger X_{AA} W_A$. Its trace is invariant, whereas its individual diagonal elements generally are not. The physical projector $P = \sum_{n\in A}|u_n\rangle\langle u_n|$ is unchanged by rotations confined to A. The matrices $\widetilde{F}$, $\widetilde{G}$, and $\widetilde{\Omega}$ are covariant, and their subspace traces are invariant. With occupations or other energy-dependent weights, the invariant object is $\mathrm{Tr}_A[f(\mathcal{H}_{AA})X_{AA}]$; both factors must be rotated. In matrix elements,

$$\mathrm{Tr}_A\, X = \sum_{n\in A} X_{nn}, \qquad \mathrm{Tr}_A[f(\mathcal{H}_{AA})X_{AA}] = \sum_{m,n\in A}[f(\mathcal{H}_{AA})]_{mn}X_{nm}. \tag{34}$$

Rotations mixing A with B change the chosen active projector and are outside this fixed-subspace covariance statement.

D. Comma, ordinary, and covariant derivatives

For a specified Wannier-gauge matrix X^W, the comma derivative means

$$X^{,d} = U^\dagger(\partial_d X^W)U, \qquad \partial_d \bar{X} = X^{,d} + \bar{X}D^d - D^d\bar{X}. \tag{35}$$

The matrix-element version uses primed intermediate states in Full:

$$\begin{aligned} X^{,d}_{jk} &= (U^\dagger)_{jj'}(\partial_d X^W_{j'k'})U_{k'k}, \\ \partial_d \bar{X}_{jk} &= X^{,d}_{jk} + (\bar{X}_{jj'}D^d_{j'k} - D^d_{jj'}\bar{X}_{j'k}). \end{aligned} \tag{36}$$

Thus $X^{\cdot,d}$ is not the ordinary derivative of $\bar{X}$. In a fixed basis for an abstract operator it becomes $\langle j|\partial_d \hat{X}|k\rangle$. Such an operator-derivative matrix is covariant; it is the ordinary derivative of a transformed matrix that acquires derivatives of the gauge rotation.

Under $U \mapsto U\,\mathrm{diag}(W_A, W_B)$, $X_{AB} \mapsto W_A^\dagger X_{AB} W_B$. Using the coefficient-space connection, define [1]

$$X^{:d}_{\alpha\beta} = \partial_d X_{\alpha\beta} + D^d_{\alpha\alpha} X_{\alpha\beta} - X_{\alpha\beta} D^d_{\beta\beta}, \quad \alpha,\beta \in \{A,B\}. \tag{37}$$

For each choice of row and column space this is

$$\begin{aligned}
X^{:d}_{mn} &= \partial_d X_{mn} + D^d_{mn'} X_{n'n} - X_{mn'} D^d_{n'n},\\
X^{:d}_{ml} &= \partial_d X_{ml} + D^d_{mn'} X_{n'l} - X_{ml'} D^d_{l'l},\\
X^{:d}_{ln} &= \partial_d X_{ln} + D^d_{lp'} X_{p'n} - X_{lm'} D^d_{m'n},\\
X^{:d}_{lp} &= \partial_d X_{lp} + D^d_{ll'} X_{l'p} - X_{ll'} D^d_{l'p}.
\end{aligned} \tag{38}$$

Here α,β label blocks, while the superscript d labels a Cartesian direction. The symbol X in this equation denotes a matrix in the chosen band frame, which may be the Hamiltonian gauge or a block rotation of it. For a rotated Wannier matrix the resulting block rules are

$$X^{:d}_{AB} = X^{,d}_{AB} - D^d_{AB} X_{BB} + X_{AA} D^d_{AB}, \tag{39}$$

$$X^{:d}_{BA} = X^{,d}_{BA} - D^d_{BA} X_{AA} + X_{BB} D^d_{BA}, \tag{40}$$

$$X^{:d}_{AA} = X^{,d}_{AA} - D^d_{AB} X_{BA} + X_{AB} D^d_{BA}. \tag{41}$$

In matrix elements, these interpolation rules become

$$\begin{aligned}
X^{:d}_{ml} &= X^{,d}_{ml} - D^d_{ml'} X_{l'l} + X_{mn'} D^d_{n'l},\\
X^{:d}_{ln} &= X^{,d}_{ln} - D^d_{lm'} X_{m'n} + X_{lp'} D^d_{p'n},\\
X^{:d}_{mn} &= X^{,d}_{mn} - D^d_{ml'} X_{l'n} + X_{ml'} D^d_{l'n},\\
X^{:d}_{lp} &= X^{,d}_{lp} - D^d_{lm'} X_{m'p} + X_{lm'} D^d_{m'p}.
\end{aligned} \tag{42}$$

In particular, $\mathcal{H}^{:d}_{mn} = \mathcal{H}^{,d}_{mn}$, $\mathcal{H}^{:d}_{lp} = \mathcal{H}^{,d}_{lp}$, and $\mathcal{H}^{:d}_{ml} = \mathcal{H}^{:d}_{ln} = 0$.

For the Hamiltonian, $\mathcal{H}^{:d}_{AA} = \mathcal{H}^{,d}_{AA}$ and $\mathcal{H}^{:d}_{AB} = 0$. The colon derivative is covariant under band-frame rotations at fixed Wannier representation:

$$(X'_{\alpha\beta})^{:a} = W_\alpha^\dagger X^{:a}_{\alpha\beta} W_\beta. \tag{43}$$

For example,

$$(X'_{mn})^{:a} = (W^\dagger)_{mm'} X^{:a}_{m'n'} W_{n'n}, \qquad (X'_{ml})^{:a} = (W^\dagger)_{mm'} X^{:a}_{m'l'} W_{l'l}. \tag{44}$$

Its definition uses D, rather than the full physical Bloch connection. To distinguish the latter construction, write

$$\begin{aligned}
\nabla_a X_{\alpha\beta} &\equiv \partial_a X_{\alpha\beta} - \mathrm{i} A^{H,a}_{\alpha\alpha} X_{\alpha\beta} + \mathrm{i} X_{\alpha\beta} A^{H,a}_{\beta\beta}\\
&= X^{:a}_{\alpha\beta} - \mathrm{i}\bar{A}^a_{\alpha\alpha} X_{\alpha\beta} + \mathrm{i} X_{\alpha\beta} \bar{A}^a_{\beta\beta}.
\end{aligned} \tag{45}$$

Its matrix elements in the four index spaces are

$$\begin{aligned}
(\nabla_a X)_{mn} &= X^{:a}_{mn} - \mathrm{i}\bar{A}^a_{mn'} X_{n'n} + \mathrm{i} X_{mn'} \bar{A}^a_{n'n},\\
(\nabla_a X)_{ml} &= X^{:a}_{ml} - \mathrm{i}\bar{A}^a_{mn'} X_{n'l} + \mathrm{i} X_{ml'} \bar{A}^a_{l'l},\\
(\nabla_a X)_{ln} &= X^{:a}_{ln} - \mathrm{i}\bar{A}^a_{lp'} X_{p'n} + \mathrm{i} X_{lm'} \bar{A}^a_{m'n},\\
(\nabla_a X)_{lp} &= X^{:a}_{lp} - \mathrm{i}\bar{A}^a_{ll'} X_{l'p} + \mathrm{i} X_{ll'} \bar{A}^a_{l'p}.
\end{aligned} \tag{46}$$

Both derivatives transform covariantly under W_α, W_β, but they need not have the same matrix elements. The physical connection A^H is unchanged by the compensated Wannier-basis change in Eq. (30); D alone is not. Their difference has zero trace for an AA block, so both give $\partial_a \,\mathrm{Tr}_A\, X$ after taking that trace. The block interpolation identities in this handbook use the colon convention; replacing D by $-\mathrm{i}A^H$ in those identities would require rederiving the comma terms.

a. Abelian derivative used in the isolated-band formulae. For a matrix element transforming as $X'_{jk} = e^{\mathrm{i}(\phi_k-\phi_j)}X_{jk}$, our coefficient-space phase derivative is

$$X^{:a}_{jk} \equiv \partial_a X_{jk} + (D^a_{jj} - D^a_{kk})X_{jk}. \tag{47}$$

The physical Bloch-state version is

$$(\nabla_a X)_{jk} = \partial_a X_{jk} - \mathrm{i}(A^a_j - A^a_k)X_{jk} = X^{:a}_{jk} - \mathrm{i}(\bar{A}^a_{jj} - \bar{A}^a_{kk})X_{jk}. \tag{48}$$

They coincide in the fixed-basis model of Sec. V, where $A^W = 0$ and $A^H = \mathrm{i}D$. In a general Wannier basis, they must be distinguished. The semicolon in the BCP quotient formulae refers to Eq. (47).

b. Parallel transport is a local condition. Along a path $\boldsymbol{k}(s)$, coefficient parallel transport of a nondegenerate band means $D^{\parallel}_{nn} = \sum_a \dot{k}_a D^a_{nn} = 0$. A phase can be chosen to impose this condition along an open smooth path, or to set the diagonal connection to zero at a specified point. Physical Bloch parallel transport instead means $A^{H,\parallel}_n = \mathrm{i}\langle u_n|\mathrm{d}u_n/\mathrm{d}s\rangle = 0$. These choices differ when $\bar{A}^{\parallel}_{nn} \neq 0$. The label $\parallel$ denotes contraction with the path tangent, not a band index. Neither choice removes interband D_{jk} or A^H_{jk} $(j \neq k)$. Nonzero curvature prevents a connection from being set to zero throughout a whole patch; nontrivial holonomy can also obstruct a periodic gauge with zero connection around a closed loop. A block parallel-transport frame can similarly set its tangential intrablock connection to zero, but may mix distinct energies and leave the Hamiltonian gauge.

c. Reading the subsequent formulae. Use the fixed Wannier convention for the colon derivatives and Fourier inputs. Evaluate explicit energy denominators in the Hamiltonian gauge, using block identities when rotating the active space. Use isolated-band phase derivatives only where the relevant bands are nondegenerate. The notation $\bar{X}$ means a matrix rotation, and $\widetilde{X}$ denotes a projected construction; neither symbol defines an additional gauge.

E. Trace, product, and chain rules

The trace and product rules eliminate gauge-dependent intrablock connections:

$$\partial_d \operatorname{Tr}_A X = \operatorname{Tr}_A X^{:d}, \tag{49}$$

$$(X_{\alpha\beta}Y_{\beta\gamma})^{:d} = X^{:d}_{\alpha\beta}Y_{\beta\gamma} + X_{\alpha\beta}Y^{:d}_{\beta\gamma}. \tag{50}$$

In components, representative A- and B-space contractions are

$$\begin{aligned}
\partial_d \sum_{n\in A} X_{nn} &= \sum_{n\in A} X^{:d}_{nn}, \\
(X_{mn'}Y_{n'n})^{:d} &= X^{:d}_{mn'}Y_{n'n} + X_{mn'}Y^{:d}_{n'n}, \\
(X_{ml'}Y_{l'n})^{:d} &= X^{:d}_{ml'}Y_{l'n} + X_{ml'}Y^{:d}_{l'n}.
\end{aligned} \tag{51}$$

Equation (50) uses fixed blocks $\alpha, \beta, \gamma \in \{A, B\}$; the product contracts state indices inside the intermediate block β, with no implicit sum over blocks. For $\widetilde{f} = f(\mathcal{H})$, evaluation in the energy eigenbasis gives

$$\widetilde{f}^{:d}_{mn} = \mathcal{H}^{:d}_{mn} \begin{cases} f'(\varepsilon_n), & \varepsilon_m = \varepsilon_n, \\ \dfrac{f(\varepsilon_m) - f(\varepsilon_n)}{\varepsilon_m - \varepsilon_n}, & \varepsilon_m \neq \varepsilon_n. \end{cases} \tag{52}$$

The divided difference is essential when a block contains distinct energies. At zero temperature, apply the smooth-function rule before taking the appropriate distributional limit of the occupation.

For compact second derivatives, define the bilinear product expansion

$$\begin{aligned}
\mathscr{P}^d(X,Y) &= X^{:d}Y + XY^{:d}, \\
\mathscr{P}^{de}(X,Y) &= X^{:de}Y + X^{:d}Y^{:e} + X^{:e}Y^{:d} + XY^{:de}, \\
X^{:de} &\equiv (X^{:d})^{:e}.
\end{aligned} \tag{53}$$

For an intermediate state in B, its fully indexed second-order form is

$$[\mathscr{P}^{de}(X_{AB}, Y_{BA})]_{mn} = X^{:de}_{ml'}Y_{l'n} + X^{:d}_{ml'}Y^{:e}_{l'n} + X^{:e}_{ml'}Y^{:d}_{l'n} + X_{ml'}Y^{:de}_{l'n}. \tag{54}$$

For an intermediate state in A, replace $l' \in B$ by $n' \in A$ in all four terms.

Matrix order is preserved. Ordered covariant derivatives need not commute; ordinary derivatives of a smooth traced scalar do commute.

III. GAUGE-COVARIANT GEOMETRY AND WANNIER INTERPOLATION

A. Projectors and geometric matrices

Let $P = \sum_{n\in A} |u_n\rangle\langle u_n|$ and $Q = 1 - P$. For physical Bloch states, Q includes states outside a finite Wannier space. The projected geometric matrices are

$$\widetilde{F}^{ab}_{mn} = \langle \partial_a u_m | Q | \partial_b u_n \rangle, \tag{55}$$

$$\widetilde{\mathcal{E}}^{ab} = \tfrac{1}{2}(\mathcal{H}_{AA}\widetilde{F}^{ab} + \widetilde{F}^{ab}\mathcal{H}_{AA}), \tag{56}$$

$$\widetilde{H}^{ab}_{mn} = \langle \partial_a u_m | Q\hat{\mathcal{H}}Q | \partial_b u_n \rangle. \tag{57}$$

The energy-weighted matrix in Eq. (56) has elements

$$\widetilde{\mathcal{E}}^{ab}_{mn} = \tfrac{1}{2}\left(\mathcal{H}_{mn'}\widetilde{F}^{ab}_{n'n} + \widetilde{F}^{ab}_{mn'}\mathcal{H}_{n'n}\right). \tag{58}$$

Each transforms as $\widetilde{O} \mapsto W_A^\dagger \widetilde{O} W_A$ and obeys $\widetilde{O}^{ab}_{mn} = (\widetilde{O}^{ba}_{nm})^*$. In the Hamiltonian gauge, $\widetilde{\mathcal{E}}^{ab}_{mn} = (\varepsilon_m + \varepsilon_n)\widetilde{F}^{ab}_{mn}/2$. The metric and curvature matrices are

$$\widetilde{G}^{ab} = \tfrac{1}{2}(\widetilde{F}^{ab} + \widetilde{F}^{ba}), \qquad \widetilde{\Omega}^{c} = \mathrm{i}\epsilon_{cab}\widetilde{F}^{ab}. \tag{59}$$

Element by element,

$$\widetilde{G}^{ab}_{mn} = \tfrac{1}{2}(\widetilde{F}^{ab}_{mn} + \widetilde{F}^{ba}_{mn}), \qquad \widetilde{\Omega}^{c}_{mn} = \mathrm{i}\epsilon_{cab}\widetilde{F}^{ab}_{mn}. \tag{60}$$

Here $\widetilde{G}$ denotes the metric; $\widetilde{\mathcal{E}}$ in Eq. (58) remains the energy-weighted matrix.

For $A = \{n\}$ these recover Sec. I. For a larger A, $\mathrm{Tr}_A\,\widetilde{G}$ is the metric of the whole subspace; it is generally not $\sum_{n\in A} G_n$.

For a continuum Hamiltonian with kinetic energy $(\boldsymbol{p} + \hbar\boldsymbol{k})^2/(2m_e)$ and a local, momentum-independent potential, the fourth matrix in `Wannier.tex` is

$$\widetilde{S}^{ab}_{mn} = \frac{\hbar^2}{m_e}\delta_{ab}\delta_{mn} + 2(\widetilde{\mathcal{E}}^{ab}_{mn} - \widetilde{H}^{ab}_{mn}). \tag{61}$$

For $A = \{n\}$, $\mathrm{Re}\,\widetilde{S}^{ab}_{nn} = \partial_a\partial_b\varepsilon_n$. The free-electron term cannot be substituted for the bare Hessian of a general tight-binding or nonlocal Hamiltonian; use Sec. VI there.

B. Internal and external terms

Define Wannier-gauge matrices

$$\begin{aligned} F^{W,ab}_{jk} &= \langle \partial_a u^W_j | \partial_b u^W_k \rangle, \\ B^{W,a}_{jk} &= \mathrm{i}\langle u^W_j | \hat{\mathcal{H}} | \partial_a u^W_k \rangle, \\ H^{W,ab}_{jk} &= \langle \partial_a u^W_j | \hat{\mathcal{H}} | \partial_b u^W_k \rangle. \end{aligned} \tag{62}$$

The two-index $H^{W,ab}$ is an energy-weighted geometric matrix, not $\partial_a\partial_b\mathcal{H}^W$. Then

$$\widetilde{F}^{ab}_{\mathrm{int}} = -D^a_{AB}D^b_{BA}, \tag{63}$$

$$\widetilde{F}^{ab}_{\mathrm{ext}} = \bar{F}^{ab}_{AA} - \bar{A}^a_{AA}\bar{A}^b_{AA} + \mathrm{i}D^a_{AB}\bar{A}^b_{BA} + \mathrm{i}\bar{A}^a_{AB}D^b_{BA}, \tag{64}$$

$$\widetilde{F}^{ab} = \widetilde{F}^{ab}_{\mathrm{int}} + \widetilde{F}^{ab}_{\mathrm{ext}}. \tag{65}$$

The same internal and external terms, with every band index shown, are

$$\begin{aligned} \widetilde{F}^{ab}_{mn,\mathrm{int}} &= -D^a_{ml'}D^b_{l'n}, \\ \widetilde{F}^{ab}_{mn,\mathrm{ext}} &= \bar{F}^{ab}_{mn} - \bar{A}^a_{mn'}\bar{A}^b_{n'n} + \mathrm{i}D^a_{ml'}\bar{A}^b_{l'n} + \mathrm{i}\bar{A}^a_{ml'}D^b_{l'n}, \\ \widetilde{F}^{ab}_{mn} &= \widetilde{F}^{ab}_{mn,\mathrm{int}} + \widetilde{F}^{ab}_{mn,\mathrm{ext}}. \end{aligned} \tag{66}$$

Thus the DD and mixed products sum over $l' \in B$, while the $\bar{A}\bar{A}$ subtraction sums over $n' \in A$.

Here B is the complement of A within the represented Wannier space; $\bar{F}$ retains the additional basis-derivative information. Keeping only $\widetilde{F}_{\rm int}$ corresponds to the fixed-basis Hamiltonian geometry.

For $\widetilde{H}$, assuming the represented active states are eigenstates of the physical Hamiltonian (or lie in the exact frozen subspace),

$$\begin{aligned}
\widetilde{H}^{ab}_{\rm int} &= -D^a_{AB}\mathcal{H}_{BB}D^b_{BA},\\
\widetilde{H}^{ab}_{\rm ext} &= \bar{H}^{ab}_{AA} - \bar{A}^a_{AA}\mathcal{H}_{AA}\bar{A}^b_{AA} + \mathrm{i}D^a_{AB}\bar{B}^b_{BA} + \mathrm{i}(\bar{B}^a)^\dagger_{AB}D^b_{BA}.
\end{aligned} \tag{67}$$

The corresponding matrix elements are

$$\begin{aligned}
\widetilde{H}^{ab}_{mn,\rm int} &= -D^a_{ml'}\mathcal{H}_{l'p'}D^b_{p'n},\\
\widetilde{H}^{ab}_{mn,\rm ext} &= \bar{H}^{ab}_{mn} - \bar{A}^a_{mn'}\mathcal{H}_{n'n''}\bar{A}^b_{n''n} + \mathrm{i}D^a_{ml'}\bar{B}^b_{l'n} + \mathrm{i}(\bar{B}^a_{l'm})^*D^b_{l'n}.
\end{aligned} \tag{68}$$

In the Hamiltonian gauge the first line is $\widetilde{H}^{ab}_{mn,\rm int} = -D^a_{ml'}\varepsilon_{l'}D^b_{l'n}$, and the subtracted external product is $-\bar{A}^a_{mn'}\varepsilon_{n'}\bar{A}^b_{n'n}$.

Outside that exact subspace, projection and disentanglement errors must be assessed before interpreting the interpolation as a physical matrix element.

A useful sign check is

$$\Omega^{ab}_{A,\rm int} = 2\,\mathrm{Im}\,\mathrm{Tr}_A(D^a_{AB}D^b_{BA}), \qquad \widetilde{\Omega}^z = \mathrm{i}(\widetilde{F}^{xy} - \widetilde{F}^{yx}). \tag{69}$$

Equivalently,

$$\Omega^{ab}_{A,\rm int} = 2\,\mathrm{Im}\sum_{n\in A}\sum_{l\in B} D^a_{nl}D^b_{ln}, \qquad \widetilde{\Omega}^z_{mn} = \mathrm{i}(\widetilde{F}^{xy}_{mn} - \widetilde{F}^{yx}_{mn}). \tag{70}$$

C. First and second geometric derivatives

The internal first derivative is

$$\widetilde{F}^{ab:d}_{\rm int} = -D^{a:d}_{AB}D^b_{BA} - D^a_{AB}D^{b:d}_{BA}. \tag{71}$$

The external first derivative contains all seven product-rule terms:

$$\begin{aligned}
\widetilde{F}^{ab:d}_{\rm ext} = \bar{F}^{ab:d}_{AA} &- \bar{A}^{a:d}_{AA}\bar{A}^b_{AA} - \bar{A}^a_{AA}\bar{A}^{b:d}_{AA}\\
&+ \mathrm{i}D^{a:d}_{AB}\bar{A}^b_{BA} + \mathrm{i}D^a_{AB}\bar{A}^{b:d}_{BA} + \mathrm{i}\bar{A}^{a:d}_{AB}D^b_{BA} + \mathrm{i}\bar{A}^a_{AB}D^{b:d}_{BA}.
\end{aligned} \tag{72}$$

For implementation, the explicit first-derivative matrix elements are

$$\begin{aligned}
\widetilde{F}^{ab:d}_{mn,\rm int} &= -D^{a:d}_{ml'}D^b_{l'n} - D^a_{ml'}D^{b:d}_{l'n},\\
\widetilde{F}^{ab:d}_{mn,\rm ext} &= \bar{F}^{ab:d}_{mn} - \bar{A}^{a:d}_{mn'}\bar{A}^b_{n'n} - \bar{A}^a_{mn'}\bar{A}^{b:d}_{n'n}\\
&\quad + \mathrm{i}D^{a:d}_{ml'}\bar{A}^b_{l'n} + \mathrm{i}D^a_{ml'}\bar{A}^{b:d}_{l'n} + \mathrm{i}\bar{A}^{a:d}_{ml'}D^b_{l'n} + \mathrm{i}\bar{A}^a_{ml'}D^{b:d}_{l'n}.
\end{aligned} \tag{73}$$

The derivative does not change the intermediate-state space: the two $\bar{A}\bar{A}$ terms still contract $n' \in A$, not $l' \in B$.

All second-order terms are retained compactly by Eq. (53):

$$\begin{aligned}
\widetilde{F}^{ab:de}_{\rm int} &= -\mathscr{P}^{de}(D^a_{AB}, D^b_{BA}),\\
\widetilde{F}^{ab:de}_{\rm ext} &= \bar{F}^{ab:de}_{AA} - \mathscr{P}^{de}(\bar{A}^a_{AA}, \bar{A}^b_{AA})\\
&\quad + \mathrm{i}\mathscr{P}^{de}(D^a_{AB}, \bar{A}^b_{BA}) + \mathrm{i}\mathscr{P}^{de}(\bar{A}^a_{AB}, D^b_{BA}).
\end{aligned} \tag{74}$$

Expanding the second derivative gives

$$\begin{aligned}
\widetilde{F}^{ab:de}_{mn,\rm int} &= -D^{a:de}_{ml'}D^b_{l'n} - D^{a:d}_{ml'}D^{b:e}_{l'n} - D^{a:e}_{ml'}D^{b:d}_{l'n} - D^a_{ml'}D^{b:de}_{l'n},\\
\widetilde{F}^{ab:de}_{mn,\rm ext} &= \bar{F}^{ab:de}_{mn} - \bar{A}^{a:de}_{mn'}\bar{A}^b_{n'n} - \bar{A}^{a:d}_{mn'}\bar{A}^{b:e}_{n'n}\\
&\quad - \bar{A}^{a:e}_{mn'}\bar{A}^{b:d}_{n'n} - \bar{A}^a_{mn'}\bar{A}^{b:de}_{n'n}\\
&\quad + \mathrm{i}D^{a:de}_{ml'}\bar{A}^b_{l'n} + \mathrm{i}D^{a:d}_{ml'}\bar{A}^{b:e}_{l'n} + \mathrm{i}D^{a:e}_{ml'}\bar{A}^{b:d}_{l'n} + \mathrm{i}D^a_{ml'}\bar{A}^{b:de}_{l'n}\\
&\quad + \mathrm{i}\bar{A}^{a:de}_{ml'}D^b_{l'n} + \mathrm{i}\bar{A}^{a:d}_{ml'}D^{b:e}_{l'n} + \mathrm{i}\bar{A}^{a:e}_{ml'}D^{b:d}_{l'n} + \mathrm{i}\bar{A}^a_{ml'}D^{b:de}_{l'n}.
\end{aligned} \tag{75}$$

Consequently,

$$\widetilde{G}^{ab:d} = \tfrac{1}{2}(\widetilde{F}^{ab:d} + \widetilde{F}^{ba:d}), \qquad \widetilde{\Omega}^{c:d} = \mathrm{i}\epsilon_{cab}\widetilde{F}^{ab:d},$$
$$\widetilde{G}^{ab:de} = \tfrac{1}{2}(\widetilde{F}^{ab:de} + \widetilde{F}^{ba:de}), \qquad \widetilde{\Omega}^{c:de} = \mathrm{i}\epsilon_{cab}\widetilde{F}^{ab:de}. \tag{76}$$

The metric and curvature derivatives are obtained without suppressing band indices:

$$\widetilde{G}^{ab:d}_{mn} = \tfrac{1}{2}(\widetilde{F}^{ab:d}_{mn} + \widetilde{F}^{ba:d}_{mn}), \qquad \widetilde{\Omega}^{c:d}_{mn} = \mathrm{i}\epsilon_{cab}\widetilde{F}^{ab:d}_{mn},$$
$$\widetilde{G}^{ab:de}_{mn} = \tfrac{1}{2}(\widetilde{F}^{ab:de}_{mn} + \widetilde{F}^{ba:de}_{mn}), \qquad \widetilde{\Omega}^{c:de}_{mn} = \mathrm{i}\epsilon_{cab}\widetilde{F}^{ab:de}_{mn}. \tag{77}$$

For example, the metric dipole of the active subspace is $\partial_d \sum_{n\in A} \widetilde{G}^{ab}_{nn} = \sum_{n\in A} \widetilde{G}^{ab:d}_{nn}$.

Taking Tr_A gives the ordinary first or second derivative of the corresponding subspace trace. The same product rule differentiates Eqs. (56) and (67); derivatives of every Hamiltonian factor are required.

IV. BCP AND ENERGY-DENOMINATOR DERIVATIVES

A. A symmetric block construction

For off-block elements ($n \in A$, $l \in B$), define M by Eq. (19) and

$$N^a_{nl} = \frac{M^a_{nl}}{\Delta_{nl}}, \qquad (M^a_{nl})^* = M^a_{ln}, \qquad (N^a_{nl})^* = -N^a_{ln}. \tag{78}$$

The finite-model energy-weighted tensor and BCP are

$$\widetilde{\mathcal{F}}^{ab} = \tfrac{1}{2}(N^a_{AB}M^b_{BA} - M^a_{AB}N^b_{BA}), \tag{79}$$
$$\widetilde{\mathcal{G}}^{ab} = \tfrac{1}{2}(\widetilde{\mathcal{F}}^{ab} + \widetilde{\mathcal{F}}^{ba}). \tag{80}$$

In matrix elements,

$$\widetilde{\mathcal{F}}^{ab}_{mn} = \tfrac{1}{2}(N^a_{ml'}M^b_{l'n} - M^a_{ml'}N^b_{l'n}),$$
$$\widetilde{\mathcal{G}}^{ab}_{mn} = \tfrac{1}{2}(\widetilde{\mathcal{F}}^{ab}_{mn} + \widetilde{\mathcal{F}}^{ba}_{mn}). \tag{81}$$

In the energy eigenbasis,

$$\widetilde{\mathcal{F}}^{ab}_{mn} = \frac{1}{2}\sum_{l\in B} M^a_{ml}M^b_{ln}\left(\frac{1}{\Delta_{ml}} + \frac{1}{\Delta_{nl}}\right). \tag{82}$$

For a single active band and a complete model complement, $\widetilde{\mathcal{G}}^{ab}_{nn} = \mathcal{G}^{ab}_n$. A calculation using only M within a truncated Wannier space does not restore energy-denominator contributions from all excluded physical bands.

Let $T^{ab} = N^a_{AB}M^b_{BA} - M^a_{AB}N^b_{BA}$. Hermiticity gives $(T^{ab})^\dagger = T^{ba}$, so the code-friendly form is

$$\widetilde{\mathcal{G}}^{ab} = \tfrac{1}{4}[T^{ab} + (T^{ab})^\dagger]. \tag{83}$$

Here the Hermitian symmetrization is explicitly

$$T^{ab}_{mn} = N^a_{ml'}M^b_{l'n} - M^a_{ml'}N^b_{l'n}, \qquad \widetilde{\mathcal{G}}^{ab}_{mn} = \tfrac{1}{4}[T^{ab}_{mn} + (T^{ab}_{nm})^*]. \tag{84}$$

This explains the division by four after band-index Hermitian symmetrization in the archived `BCP` class.

B. Connection derivatives and the general block rule

The first derivative of the off-block coefficient connection is

$$D^{a:d}_{ln} = \frac{1}{\varepsilon_n - \varepsilon_l}\left[\mathcal{H}^{:ad}_{ln} + \mathcal{H}^{:a}_{lp'}D^d_{p'n} - D^d_{lm'}\mathcal{H}^{:a}_{m'n} + \mathcal{H}^{:d}_{lp'}D^a_{p'n} - D^a_{lm'}\mathcal{H}^{:d}_{m'n}\right], \tag{85}$$

where $m' \in A$ and $p' \in B$ are summed. Also $M^{a:d} = \bar{A}^{a:d} + \mathrm{i}D^{a:d}$, with the corresponding relation at second order.

To differentiate an energy denominator covariantly, define the Sylvester operator

$$\mathscr{L}_{\alpha\beta}(Y) = \mathcal{H}_{\alpha\alpha}Y - Y\mathcal{H}_{\beta\beta}. \tag{86}$$

Here $\alpha \neq \beta$, and Y, R, N^a, M^a in the Sylvester equations denote $\alpha\beta$ blocks. Its inverse exists for separated spectra. Equation (78) is the Hamiltonian-gauge evaluation of $\mathscr{L}_{\alpha\beta}(N^a) = M^a$. For $\mathscr{L}_{\alpha\beta}(Y) = R$, product differentiation gives

$$\mathscr{L}_{\alpha\beta}(Y^{:d}) = R^{:d} - \mathcal{H}^{:d}_{\alpha\alpha}Y + Y\mathcal{H}^{:d}_{\beta\beta}, \tag{87}$$

$$\mathscr{L}_{\alpha\beta}(Y^{:de}) = R^{:de} - \mathcal{H}^{:de}_{\alpha\alpha}Y - \mathcal{H}^{:d}_{\alpha\alpha}Y^{:e} - \mathcal{H}^{:e}_{\alpha\alpha}Y^{:d} + Y^{:d}\mathcal{H}^{:e}_{\beta\beta} + Y^{:e}\mathcal{H}^{:d}_{\beta\beta} + Y\mathcal{H}^{:de}_{\beta\beta}. \tag{88}$$

For the BA elements, the Sylvester equation and both derivatives read

$$\begin{aligned}
&\mathcal{H}_{lp'}Y_{p'n} - Y_{lm'}\mathcal{H}_{m'n} = R_{ln},\\
&\mathcal{H}_{lp'}Y^{:d}_{p'n} - Y^{:d}_{lm'}\mathcal{H}_{m'n} = R^{:d}_{ln} - \mathcal{H}^{:d}_{lp'}Y_{p'n} + Y_{lm'}\mathcal{H}^{:d}_{m'n},\\
&\mathcal{H}_{lp'}Y^{:de}_{p'n} - Y^{:de}_{lm'}\mathcal{H}_{m'n} = R^{:de}_{ln} - \mathcal{H}^{:de}_{lp'}Y_{p'n} - \mathcal{H}^{:d}_{lp'}Y^{:e}_{p'n} - \mathcal{H}^{:e}_{lp'}Y^{:d}_{p'n}\\
&\qquad\qquad + Y^{:d}_{lm'}\mathcal{H}^{:e}_{m'n} + Y^{:e}_{lm'}\mathcal{H}^{:d}_{m'n} + Y_{lm'}\mathcal{H}^{:de}_{m'n}.
\end{aligned} \tag{89}$$

Thus the sums over $p' \in B$ and $m' \in A$ stay distinct even after repeated differentiation. The AB equation has elements

$$\mathcal{H}_{mn'}Y_{n'l} - Y_{ml'}\mathcal{H}_{l'l} = R_{ml}; \tag{90}$$

its derivatives follow the same product rule, with these index spaces. For BCP use $Y_{ln} = N^a_{ln}$, $R_{ln} = M^a_{ln}$; for the coefficient connection use $Y_{ln} = D^a_{ln}$, $R_{ln} = -\mathcal{H}^a_{ln}$. In the Hamiltonian gauge the left side of the first line becomes $(\varepsilon_l - \varepsilon_n)Y_{ln}$.

Set $(Y, R) = (N^a, M^a)$ for BCP denominators, or $(Y, R) = (D^a_{BA}, -\mathcal{H}^a_{BA})$ for coefficient connections. These identities preserve intrablock matrix products and supply a general reference for auditing the multiband implementation.

C. Isolated-band quotient formulae

For individual nondegenerate eigenlines, use the coefficient-space phase-covariant derivative $;d$ defined in Eq. (47). It differs from a multiband block derivative and, in a moving Wannier basis, from the physical Bloch derivative in Eq. (48). Let $\Delta^{;d}_{jk} = E^d_j - E^d_k$ and $\Delta^{;de}_{jk} = \partial_e\partial_d(\varepsilon_j - \varepsilon_k)$. The scalar comma notation here is simply $\Delta^{;d}_{jk} = \partial_d\Delta_{jk}$ (and similarly at second order), with $j \neq k$ and nonzero Δ_{jk}. Then

$$N^{a;d}_{jk} = \frac{M^{a;d}_{jk}}{\Delta_{jk}} - \frac{M^a_{jk}\Delta^{;d}_{jk}}{\Delta^2_{jk}}, \tag{91}$$

$$N^{a;de}_{jk} = \frac{M^{a;de}_{jk}}{\Delta_{jk}} - \frac{M^{a;d}_{jk}\Delta^{;e}_{jk} + M^{a;e}_{jk}\Delta^{;d}_{jk} + M^a_{jk}\Delta^{;de}_{jk}}{\Delta^2_{jk}} + \frac{2M^a_{jk}\Delta^{;d}_{jk}\Delta^{;e}_{jk}}{\Delta^3_{jk}}. \tag{92}$$

These are the compact quotient formulae in `Wannier.tex`. They must not replace Eqs. (87)–(88) when the numerator is evaluated with a multiband colon derivative. The archived `DerBCP` mixes these two prescriptions: it uses the multiband derivative of M, but only diagonal energy gradients in the denominator derivative. The missing terms are given explicitly below.

D. Missing denominator terms in the archived DerBCP

a. The exact matrix-element rule. Evaluate Eq. (87) in the Hamiltonian gauge, after taking the covariant derivative. For $m, n \in A$ and $l, p \in B$, it gives

$$\begin{aligned}
N^{a:d}_{ml} &= \frac{M^{a:d}_{ml} - \mathcal{H}^{:d}_{mn'}N^a_{n'l} + N^a_{ml'}\mathcal{H}^{:d}_{l'l}}{\Delta_{ml}},\\
N^{a:d}_{ln} &= \frac{M^{a:d}_{ln} - \mathcal{H}^{:d}_{lp'}N^a_{p'n} + N^a_{lm'}\mathcal{H}^{:d}_{m'n}}{\Delta_{ln}}.
\end{aligned} \tag{93}$$

Primed indices are implicitly summed in their assigned spaces. Thus the first line contains both an A-space matrix product and a B-space matrix product, even when the unperturbed Hamiltonian is diagonal.

By contrast, extracting only the diagonal entries of the Hamiltonian derivative, as in the archived `Vn`, `Vl`, and `deltaV` operations, gives the estimates

$$\widehat{N}^{a:d}_{ml} = \frac{M^{a:d}_{ml}}{\Delta_{ml}} - \frac{N^{a}_{ml}(E^{d}_{m} - E^{d}_{l})}{\Delta_{ml}},$$
$$\widehat{N}^{a:d}_{ln} = \frac{M^{a:d}_{ln}}{\Delta_{ln}} - \frac{N^{a}_{ln}(E^{d}_{l} - E^{d}_{n})}{\Delta_{ln}}, \qquad E^{d}_{j} = \partial_{d}\varepsilon_{j}. \tag{94}$$

The hat denotes an incomplete estimate of the colon derivative, not a different gauge choice. The scalar quotient rule (91) remains correct for consistently evaluated Abelian semicolon derivatives; the error is applying its denominator term to a multiband colon derivative of the numerator.

To expose the omission without suppressing any implicit sums, define the off-diagonal parts within each space, in the Hamiltonian gauge,

$$\mathcal{H}^{:d}_{mn,\mathrm{off}} = \mathcal{H}^{:d}_{mn} - \delta_{mn}E^{d}_{m}, \qquad \mathcal{H}^{:d}_{lp,\mathrm{off}} = \mathcal{H}^{:d}_{lp} - \delta_{lp}E^{d}_{l}. \tag{95}$$

There is no sum over the unprimed indices in this definition. The omitted contributions $\delta N^{a:d} = N^{a:d} - \widehat{N}^{a:d}$ are exactly

$$\delta N^{a:d}_{ml} = \frac{-\mathcal{H}^{:d}_{mn',\mathrm{off}}N^{a}_{n'l} + N^{a}_{ml'}\mathcal{H}^{:d}_{l'l,\mathrm{off}}}{\Delta_{ml}},$$
$$\delta N^{a:d}_{ln} = \frac{-\mathcal{H}^{:d}_{lp',\mathrm{off}}N^{a}_{p'n} + N^{a}_{lm'}\mathcal{H}^{:d}_{m'n,\mathrm{off}}}{\Delta_{ln}}. \tag{96}$$

For a smooth nondegenerate eigenframe, the colon definition itself shows why these terms need not vanish:

$$\mathcal{H}^{:d}_{mn} = \delta_{mn}E^{d}_{m} + (\varepsilon_{n} - \varepsilon_{m})D^{d}_{mn} = \mathcal{H}^{,d}_{mn}. \tag{97}$$

Diagonalizing $\mathcal{H}$ therefore does not diagonalize its covariant derivative. A parallel-transport frame can remove the intragroup connection at a point, but then the ordinary derivative of the Hamiltonian matrix retains the same off-diagonal information.

Keeping the numerator derivatives fixed, the resulting correction to the BCP derivative is

$$\delta T^{ab:d}_{mn} = \delta N^{a:d}_{ml'}M^{b}_{l'n} - M^{a}_{ml'}\delta N^{b:d}_{l'n},$$
$$\delta\widetilde{\mathcal{G}}^{ab:d}_{mn} = \tfrac{1}{4}\big[\delta T^{ab:d}_{mn} + (\delta T^{ab:d}_{nm})^{*}\big]. \tag{98}$$

Hermitian symmetrization does not in general remove this correction, nor does taking the A-space trace. If both A and B contain just one band, the omitted sums vanish; this explains why a two-band Haldane check alone does not detect the error. Choosing a single active band $A = \{n\}$ is insufficient in a larger model: the off-diagonal B-space terms remain. More generally, the diagonal estimate is sufficient only when the correction in Eq. (96) vanishes for the chosen matrix elements and derivative direction.

b. Why the Sylvester operator resolves the omission. The energy denominator is the eigenbasis representation of an inverse linear operator on matrices, $N^{a} = \mathscr{L}^{-1}_{\alpha\beta}(M^{a})$. Its covariant derivative acts on both the input and the operator:

$$N^{a:d} = \mathscr{L}^{-1}_{\alpha\beta}\big[M^{a:d} - \mathscr{L}^{:d}_{\alpha\beta}(N^{a})\big],$$
$$\mathscr{L}^{:d}_{\alpha\beta}(Y) \equiv \mathcal{H}^{:d}_{\alpha\alpha}Y - Y\mathcal{H}^{:d}_{\beta\beta}. \tag{99}$$

Here $\mathscr{L}^{:d}$ differentiates the operator coefficients while holding its argument fixed. Equivalently, $(\mathscr{L}^{-1})^{:d} = -\mathscr{L}^{-1}\mathscr{L}^{:d}\mathscr{L}^{-1}$. The two Hamiltonian-derivative products produce all the terms in Eq. (93); replacing them by diagonal energy gradients discards precisely Eq. (96).

This formulation is also covariant under independent rotations within A and B. With $\mathcal{H}'_{\alpha\alpha} = W^{\dagger}_{\alpha}\mathcal{H}_{\alpha\alpha}W_{\alpha}$,

$$\mathscr{L}'_{\alpha\beta}(W^{\dagger}_{\alpha}YW_{\beta}) = W^{\dagger}_{\alpha}\mathscr{L}_{\alpha\beta}(Y)W_{\beta}. \tag{100}$$

For example, $Y'_{ln} = (W^{\dagger})_{ll'}Y_{l'm'}W_{m'n}$ and the right-hand side R_{ln} transforms by the same matrix-element rule. Uniqueness of the solution then preserves this covariance for N and its colon derivatives. Inverting $\mathscr{L}$ requires

disjoint A and B spectra, not nondegeneracy within either space; a vanishing cross-space gap must instead be treated by regrouping the bands or specifying another model.

In practice no large Sylvester solver is necessary in the Hamiltonian gauge: form the complete right-hand side with ordinary matrix products, then divide its ml or ln element by the corresponding cross-space energy difference. This is the rule used by `calculators_collect/formula/covariant_basic.py`. Differentiating the same operator equation again yields Eq. (88), including both mixed first-derivative products and the second Hamiltonian derivative. Thus the same construction fixes the denominator treatment in `DerBCP` and supplies a consistent one for `Der2BCP`.

E. BCP dipole and quadrupole kernels

With either consistently evaluated block derivatives or isolated-band derivatives,

$$\widetilde{\mathcal{F}}^{ab:d} = \tfrac{1}{2}\big[N_{AB}^{a:d}M_{BA}^{b} + N_{AB}^{a}M_{BA}^{b:d} - M_{AB}^{a:d}N_{BA}^{b} - M_{AB}^{a}N_{BA}^{b:d}\big], \tag{101}$$

$$\widetilde{\mathcal{F}}^{ab:de} = \tfrac{1}{2}\big[\mathscr{P}^{de}(N_{AB}^{a}, M_{BA}^{b}) - \mathscr{P}^{de}(M_{AB}^{a}, N_{BA}^{b})\big], \tag{102}$$

$$\widetilde{\mathcal{G}}^{ab:d} = \tfrac{1}{2}(\widetilde{\mathcal{F}}^{ab:d} + \widetilde{\mathcal{F}}^{ba:d}),$$

$$\widetilde{\mathcal{G}}^{ab:de} = \tfrac{1}{2}(\widetilde{\mathcal{F}}^{ab:de} + \widetilde{\mathcal{F}}^{ba:de}). \tag{103}$$

The first and second derivatives in matrix-element form are

$$\begin{aligned}
\widetilde{\mathcal{F}}_{mn}^{ab:d} &= \tfrac{1}{2}\big[N_{ml'}^{a:d}M_{l'n}^{b} + N_{ml'}^{a}M_{l'n}^{b:d} - M_{ml'}^{a:d}N_{l'n}^{b} - M_{ml'}^{a}N_{l'n}^{b:d}\big],\\
\widetilde{\mathcal{F}}_{mn}^{ab:de} &= \tfrac{1}{2}\big[N_{ml'}^{a:de}M_{l'n}^{b} + N_{ml'}^{a:d}M_{l'n}^{b:e} + N_{ml'}^{a:e}M_{l'n}^{b:d} + N_{ml'}^{a}M_{l'n}^{b:de}\\
&\qquad - M_{ml'}^{a:de}N_{l'n}^{b} - M_{ml'}^{a:d}N_{l'n}^{b:e} - M_{ml'}^{a:e}N_{l'n}^{b:d} - M_{ml'}^{a}N_{l'n}^{b:de}\big],\\
\widetilde{\mathcal{G}}_{mn}^{ab:d} &= \tfrac{1}{2}(\widetilde{\mathcal{F}}_{mn}^{ab:d} + \widetilde{\mathcal{F}}_{mn}^{ba:d}),\\
\widetilde{\mathcal{G}}_{mn}^{ab:de} &= \tfrac{1}{2}(\widetilde{\mathcal{F}}_{mn}^{ab:de} + \widetilde{\mathcal{F}}_{mn}^{ba:de}).
\end{aligned} \tag{104}$$

The two factors of 1/2 in this construction are independent. For a smooth subspace, $\mathrm{Tr}_A\,\widetilde{\mathcal{G}}^{ab:d}$ and $\mathrm{Tr}_A\,\widetilde{\mathcal{G}}^{ab:de}$ are the dipole and quadrupole geometric kernels before occupations and response prefactors are applied.

V. EXPLICIT BAND-SUM AND MATRIX FORMULAE FROM QGD

A. Domain and matrix ingredients

This section applies to a smooth Hamiltonian in a fixed orthonormal basis, with nondegenerate bands and nonzero denominators. It gives an independent representation of the internal geometry. Define energy-derivative matrices, rather than physical velocities,

$$V_{jk}^{a} = \langle j|\partial_a\mathcal{H}|k\rangle, \qquad \Gamma_{jk}^{ab} = \langle j|\partial_a\partial_b\mathcal{H}|k\rangle,$$

$$(K_q^a)_{jk} = \begin{cases} V_{jk}^{a}/\Delta_{jk}^{q}, & j \neq k,\\ 0, & j = k, \end{cases} \qquad (V_F^a)_{jk} = \delta_{jk}E_j^a. \tag{105}$$

Division by Δ^q is elementwise and excludes the diagonal. Here q is a positive integer power, not a band or Cartesian index. Products of K_q are ordinary matrix products over Full. In this convention $D_{jk}^{a} = -(K_1^a)_{jk}$ and $A_{jk}^{a} = -\mathrm{i}(K_1^a)_{jk}$ for $j \neq k$. Thus K_1 alone is not the interband Berry connection.

The QGT can be evaluated by either expression

$$Q_n^{ab} = \sum_{j\neq n}\frac{V_{nj}^{a}V_{jn}^{b}}{\Delta_{nj}^{2}} = -[K_1^aK_1^b]_{nn}. \tag{106}$$

Only the diagonal of the matrix product is asserted to equal the single-band QGT. Its off-diagonal entries are not the general projected matrix of Eq. (55).

B. QGD-I and QGD-II

Write $\partial_c Q_n^{ab} = \mathcal{I}_n^{ab;c} + \mathcal{J}_n^{ab;c}$. The one-sum contribution is

$$\mathcal{I}_n^{ab;c} = \sum_{j\neq n}\left[\frac{\Gamma_{nj}^{ca}V_{jn}^b + V_{nj}^a\Gamma_{jn}^{cb}}{\Delta_{nj}^2} - \frac{2V_{nj}^aV_{jn}^b(E_n^c - E_j^c)}{\Delta_{nj}^3}\right]. \tag{107}$$

The two-sum contribution is

$$\mathcal{J}_n^{ab;c} = \sum_{j\neq n}\Bigg[\sum_{k\neq n}\frac{V_{nk}^cV_{kj}^aV_{jn}^b + V_{nj}^aV_{jk}^bV_{kn}^c}{\Delta_{nj}^2\Delta_{nk}} + \sum_{k\neq j}\frac{V_{nk}^aV_{kj}^cV_{jn}^b + V_{nj}^aV_{jk}^cV_{kn}^b}{\Delta_{nj}^2\Delta_{jk}}\Bigg]. \tag{108}$$

The corresponding compact matrix expressions are

$$\mathcal{I}_n^{ab;c} = \left[\Gamma^{ca}K_2^b + K_2^a\Gamma^{cb} - 2[V_F^c, K_1^a]K_2^b\right]_{nn}, \tag{109}$$

$$\mathcal{J}_n^{ab;c} = \left[[K_1^c, V^a]K_2^b + K_2^a[K_1^c, V^b]\right]_{nn}. \tag{110}$$

No real part is taken until selecting the desired observable:

$$\partial_c G_n^{ab} = \mathrm{Re}\,\partial_c Q_n^{ab}, \qquad \partial_c\Omega_n^{ab} = -2\,\mathrm{Im}\,\partial_c Q_n^{ab}. \tag{111}$$

Equations (107)–(110) retain the one-sum/two-sum distinction in `QGD.tex` without repeating the intermediate bra-ket expansions.

C. A complete quotient identity for the second derivative

The QGQ expansions in the source notes are incomplete. For a compact check of any completed expansion, define the phase-invariant pair $Z_{nj}^{ab} = V_{nj}^aV_{jn}^b$ for $j \neq n$. Differentiating $Q_n^{ab} = \sum_{j\neq n} Z_{nj}^{ab}/\Delta_{nj}^2$ twice gives

$$\partial_d\partial_c Q_n^{ab} = \sum_{j\neq n}\Bigg[\frac{\partial_d\partial_c Z_{nj}^{ab}}{\Delta_{nj}^2} - \frac{2\left[(\partial_c Z_{nj}^{ab})\Delta_{nj}^{,d} + (\partial_d Z_{nj}^{ab})\Delta_{nj}^{,c} + Z_{nj}^{ab}\Delta_{nj}^{,cd}\right]}{\Delta_{nj}^3} + \frac{6Z_{nj}^{ab}\Delta_{nj}^{,c}\Delta_{nj}^{,d}}{\Delta_{nj}^4}\Bigg]. \tag{112}$$

In particular, the last coefficient is $+6$ in this scalar band-sum convention. The numerator derivatives include changes of both eigenstates and operators. For an arbitrary operator matrix X, they can be generated using

$$X_{jk}^{;c} = \langle j|\partial_c\hat{X}|k\rangle + [K_1^c, X]_{jk}, \qquad (K_q^a)_{jk}^{;c} = \frac{(V^a)_{jk}^{;c}}{\Delta_{jk}^q} - q\frac{V_{jk}^a\Delta_{jk}^{,c}}{\Delta_{jk}^{q+1}}, \quad j\neq k. \tag{113}$$

Here $;c$ removes only eigenline phase connections, with the same choice as in Sec. IV. Differentiating Γ includes $\langle j|\partial_c\partial_a\partial_b\mathcal{H}|k\rangle$. It cannot be omitted for a general tight-binding Hamiltonian. This identity is a complete differentiation rule, not a claim that the partial QGQ matrix expansions in the source notes are complete.

VI. ENERGY DERIVATIVES AND DRUDE BUILDING BLOCKS

Define the rotated bare derivatives

$$V^a = \mathcal{H}^{,a}, \quad W^{ab} = \mathcal{H}^{,ab}, \quad Y^{abc} = \mathcal{H}^{,abc}, \quad Z^{abcd} = \mathcal{H}^{,abcd}. \tag{114}$$

Here Z^{abcd} is a four-derivative Hamiltonian matrix, distinct from the band-pair numerator Z^{ab}_{nj} in Sec. V. For example,

$$V^{a:b}_{AA} = W^{ab}_{AA} + V^{a}_{AB}D^{b}_{BA} - D^{b}_{AB}V^{a}_{BA}, \tag{115}$$
$$W^{ab:c}_{AA} = Y^{abc}_{AA} + W^{ab}_{AB}D^{c}_{BA} - D^{c}_{AB}W^{ab}_{BA}, \tag{116}$$
$$V^{a:bc}_{AA} = W^{ab:c}_{AA} + V^{a:c}_{AB}D^{b}_{BA} + V^{a}_{AB}D^{b:c}_{BA} - D^{b:c}_{AB}V^{a}_{BA} - D^{b}_{AB}V^{a:c}_{BA}. \tag{117}$$

A further product derivative yields

$$\begin{aligned} V^{a:bcd}_{AA} = {} & W^{ab:cd}_{AA} + V^{a:cd}_{AB}D^{b}_{BA} + V^{a:c}_{AB}D^{b:d}_{BA} + V^{a:d}_{AB}D^{b:c}_{BA} + V^{a}_{AB}D^{b:cd}_{BA} \\ & - D^{b:cd}_{AB}V^{a}_{BA} - D^{b:c}_{AB}V^{a:d}_{BA} - D^{b:d}_{AB}V^{a:c}_{BA} - D^{b}_{AB}V^{a:cd}_{BA}. \end{aligned} \tag{118}$$

The same derivative chain, with free band indices $m, n \in A$, is

$$\begin{aligned} V^{a:b}_{mn} &= W^{ab}_{mn} + V^{a}_{ml'}D^{b}_{l'n} - D^{b}_{ml'}V^{a}_{l'n}, \\ W^{ab:c}_{mn} &= Y^{abc}_{mn} + W^{ab}_{ml'}D^{c}_{l'n} - D^{c}_{ml'}W^{ab}_{l'n}, \\ V^{a:bc}_{mn} &= W^{ab:c}_{mn} + V^{a:c}_{ml'}D^{b}_{l'n} + V^{a}_{ml'}D^{b:c}_{l'n} - D^{b:c}_{ml'}V^{a}_{l'n} - D^{b}_{ml'}V^{a:c}_{l'n}, \\ V^{a:bcd}_{mn} &= W^{ab:cd}_{mn} + V^{a:cd}_{ml'}D^{b}_{l'n} + V^{a:c}_{ml'}D^{b:d}_{l'n} + V^{a:d}_{ml'}D^{b:c}_{l'n} + V^{a}_{ml'}D^{b:cd}_{l'n} \\ & \quad - D^{b:cd}_{ml'}V^{a}_{l'n} - D^{b:c}_{ml'}V^{a:d}_{l'n} - D^{b:d}_{ml'}V^{a:c}_{l'n} - D^{b}_{ml'}V^{a:cd}_{l'n}. \end{aligned} \tag{119}$$

For a smooth isolated group,

$$\partial_a\partial_b \sum_{n\in A}\varepsilon_n = \mathrm{Tr}_A\, V^{a:b}, \quad \partial_a\partial_b\partial_c \sum_{n\in A}\varepsilon_n = \mathrm{Tr}_A\, V^{a:bc}, \quad \partial_a\partial_b\partial_c\partial_d \sum_{n\in A}\varepsilon_n = \mathrm{Tr}_A\, V^{a:bcd}. \tag{120}$$

Each trace is simply the sum of diagonal matrix elements:

$$\mathrm{Tr}_A\, V^{a:b} = \sum_{n\in A} V^{a:b}_{nn}, \qquad \mathrm{Tr}_A\, V^{a:bc} = \sum_{n\in A} V^{a:bc}_{nn}, \qquad \mathrm{Tr}_A\, V^{a:bcd} = \sum_{n\in A} V^{a:bcd}_{nn}. \tag{121}$$

For a single isolated band the inverse effective mass is $(m_n^{-1})^{ab} = \hbar^{-2}\partial_a\partial_b\varepsilon_n$. The operator Hessian W^{ab}_{nn} differs from the band-energy Hessian:

$$\partial_a\partial_b\varepsilon_n = W^{ab}_{nn} + 2\,\mathrm{Re}\sum_{j\neq n}\frac{V^{a}_{nj}V^{b}_{jn}}{\Delta_{nj}}. \tag{122}$$

Thus the orbital-quadrupole derivative of an operator Hessian must not be replaced by `Der3E`. They correspond to different objects.

VII. ORBITAL MAGNETIC QUANTITIES

A. Orbital moment and zero-temperature magnetization

With the charge convention in Eq. (4),

$$\widetilde{m}^{c} = \frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{\mathcal{E}}^{ab} - \widetilde{H}^{ab}), \tag{123}$$
$$\widetilde{\mathcal{M}}^{c}(\mu) = -\frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{H}^{ab} + \widetilde{\mathcal{E}}^{ab} - 2\mu\widetilde{F}^{ab}). \tag{124}$$

The corresponding band-resolved matrix elements are

$$\begin{aligned} \widetilde{m}^{c}_{mn} &= \frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{\mathcal{E}}^{ab}_{mn} - \widetilde{H}^{ab}_{mn}), \\ \widetilde{\mathcal{M}}^{c}_{mn}(\mu) &= -\frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{H}^{ab}_{mn} + \widetilde{\mathcal{E}}^{ab}_{mn} - 2\mu\widetilde{F}^{ab}_{mn}). \end{aligned} \tag{125}$$

The first is the symmetrized orbital-moment matrix; for an exactly degenerate active group it reduces to the usual moment matrix. For a single band, $\mathcal{M}_n^c = m_n^c + (e/\hbar)(\mu - \varepsilon_n)\Omega_n^c$. At zero temperature, using the occupied projector at each wave vector,

$$M_{\mathrm{orb}}^c = \int [\mathrm{d}\boldsymbol{k}]\, \mathrm{Tr}_{\mathrm{occ}}\, \widetilde{\mathcal{M}}^c(\mu). \tag{126}$$

Taking A to be the occupied subspace gives the explicit sum

$$M_{\mathrm{orb}}^c = \int [\mathrm{d}\boldsymbol{k}] \sum_{n\in A} \widetilde{\mathcal{M}}_{nn}^c(\mu). \tag{127}$$

At fixed μ, the derivative of Eq. (124) is

$$(\widetilde{\mathcal{M}}^c)^{;d} = -\frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{H}^{ab:d} + \widetilde{\mathcal{E}}^{ab:d} - 2\mu\widetilde{F}^{ab:d}). \tag{128}$$

At fixed μ, its matrix elements satisfy

$$\widetilde{\mathcal{M}}_{mn}^{c:d} = -\frac{\mathrm{i}e}{2\hbar}\epsilon_{cab}(\widetilde{H}_{mn}^{ab:d} + \widetilde{\mathcal{E}}_{mn}^{ab:d} - 2\mu\widetilde{F}_{mn}^{ab:d}). \tag{129}$$

Every projector derivative in $\widetilde{H}$ and $\widetilde{\mathcal{E}}$ is included through the interpolation and product rules. Keeping only $\langle\partial_a u_m|Q(\partial_d\hat{\mathcal{H}})Q|\partial_b u_n\rangle$ is insufficient.

B. Orbital magnetic quadrupole: four-term reference

To avoid the duplicated prefactors in `Wannier.tex`, this subsection uses the $e = \hbar = 1$ convention of Gao and Xiao [3]. It supplies the reduced kernel for comparison with the archive, not a new SI conversion. For an isolated band choose $A = \{n\}$ and $B = \mathrm{Full} \setminus \{n\}$, and define

$$\mathfrak{m}_{ln}^b = \frac{1}{2}\epsilon_{bcd}\sum_{p\in B}(v_{lp}^c + v_n^c\delta_{lp})A_{pn}^d, \tag{130}$$

$$\Gamma_n^{ad} = \langle u_n|\partial_a\partial_d\hat{\mathcal{H}}|u_n\rangle. \tag{131}$$

Here $v^c = \partial_c\hat{\mathcal{H}}$ in this subsection's units, and G_n^{ab} denotes the quantum metric throughout. Define four band kernels,

$$C_n^{ab} = \frac{2}{3}\,\mathrm{Re}\sum_{l\in B} A_{nl}^a\mathfrak{m}_{ln}^b, \qquad I_n^{ab} = -\frac{1}{12}\epsilon_{cdb}\partial_c\Gamma_n^{ad},$$

$$L_n^{ab} = -2\,\mathrm{Re}\sum_{l\in B}\frac{A_{nl}^a\mathfrak{m}_{ln}^b}{\Delta_{nl}}, \qquad K_n^{ab} = -\frac{1}{3}\epsilon_{cdb}\partial_c G_n^{ad}. \tag{132}$$

For the full-band sum, evaluate each isolated-band kernel above with n relabeled as j, and with its own complementary space $B = \mathrm{Full} \setminus \{j\}$. The quadrupole density is

$$\mathcal{Q}^{ab} = \int [\mathrm{d}\boldsymbol{k}] \sum_{j\in\mathrm{Full}} \left\{f_j(C_j^{ab} + I_j^{ab}) + \Phi_j(L_j^{ab} + K_j^{ab})\right\}, \tag{133}$$

where

$$f_j = \frac{1}{1 + e^{(\varepsilon_j-\mu)/(k_BT)}}, \qquad \Phi_j = -k_BT\ln\left(1 + e^{(\mu-\varepsilon_j)/(k_BT)}\right). \tag{134}$$

At $T = 0$, $f_j = \Theta(\mu - \varepsilon_j)$ and $\Phi_j = (\varepsilon_j - \mu)f_j$. The source-note labels ME1, W, ME2, QM correspond respectively to the archived v1.8 labels CL, IMD, ME, QMD.

The physical operator Hessian in Eq. (132) equals the fixed-basis matrix W of Sec. VI in that representation. With a moving Wannier basis, operator derivatives and derivatives of $\mathcal{H}^W$ require additional basis-connection terms. This distinction must be checked when comparing the archived `DerWln` implementation to the physical quadrupole formula.

The archived v1.8 calculator uses $(E - \mu)f$ for ME and QMD and defaults to `constant_factor=1`. Its weighting agrees with Eq. (134) at zero temperature, and does not by itself implement the finite-temperature grand potential. No additional $e/(2\hbar)$ factor should be inserted into Eq. (133) while retaining the convention in Eq. (130). Conversion back to physical units must be derived for the full expression.

VIII. BRIDGE TO THE V1.7 AND V1.8 CALCULATOR CATALOGUE

A. Fermi-sea and Fermi-surface kernels

For a smooth periodic band quantity X_j and vanishing boundary terms, with all band sums over $j \in \text{Full}$,

$$\int [\mathrm{d}\boldsymbol{k}] \sum_j f_j \partial_a X_j = -\int [\mathrm{d}\boldsymbol{k}] \sum_j f'_j E^a_j X_j, \tag{135}$$

$$\int [\mathrm{d}\boldsymbol{k}] \sum_j f_j \partial_a \partial_d X_j = -\int [\mathrm{d}\boldsymbol{k}] \sum_j f'_j E^a_j \partial_d X_j, \tag{136}$$

where $f'_j = \partial f/\partial\varepsilon|_{\varepsilon_j}$ is negative. A surface kernel defined with $-f'$ has the same sign as the sea integral. The code's occupation-derivative convention and outer factor must therefore be checked together. At band crossings, use smooth group traces and the matrix occupation rule rather than applying isolated-band formulae blindly.

B. Tensor axes and formula-class index

The following table records geometric kernel axes, excluding any leading wave-vector and chemical-potential axes. It is a navigation aid for the subsequent full version reconciliation.

Formula class	Geometric kernel	Raw Cartesian axes
`Metric`	G^{ab} or subspace metric	(a, b)
`DerMetric`	$G^{ab:d}$	(a, b, d)
`BCP`	$\mathcal{G}^{ab}$	(a, b)
`DerBCP`	$\mathcal{G}^{ab:d}$	(a, b, d)
`Der2BCP`	$\mathcal{G}^{ab:de}$	(a, b, d, e)
`DerOmega`, `Der2Omega`	$\Omega^{c:d}$, $\Omega^{c:de}$	(c, d), (c, d, e)
`VelBCP`	$E^c\mathcal{G}^{ab}$	(c, a, b)
`VelDerBCP`	$E^c\mathcal{G}^{ab:d}$	(c, a, b, d)
`VelDerOmega`	$E^a\Omega^{c:d}$	(a, c, d)
`InvMass`, `Der3E`, `Der4E`	band-energy derivatives	derivative order
`Qorb_GaoXiao_*`	Eq. (133)	(a, b)

For a common dipole convention (d, a, b), transpose the three axes of `DerBCP` from (a, b, d) to (d, a, b). For a common quadrupole convention (d, a, b, e), the corresponding permutation of `Der2BCP` is $(2, 0, 1, 3)$ in zero-based tensor-axis notation. Only the trailing tensor axes should be permuted when leading axes are present.

C. Items for the subsequent complete version reconciliation

The archived source and `CHANGE_REPORT.md` identify the following scope:

- v1.7 adds metric, BCP dipole/quadrupole, Berry-curvature quadrupole, fourth-energy-derivative, and early orbital-quadrupole calculators.
- v1.8 retains several formula classes but does not expose the complete v1.7 calculator set. It adds the four-term Gao–Xiao quadrupole and `Beta_GaoXiao` thermoelectric response.
- `Der2BCP_test` in v1.7 raises `NotImplementedError`. The production `Der2BCP` also carries a bug-related TODO in the source; a class name or a completed symbolic identity is not evidence of numerical validation.
- BCP denominator derivatives, active/complement choices, tensor axes, occupation signs, and SI prefactors need class-by-class comparison. Group-metric traces and sums of isolated-band metrics must be distinguished.

The thermoelectric response and the full nonlinear conductivity prefactors are outside the completed source-note consolidation here. They should be added as separate catalogue entries with their field convention, thermal kernel, component symmetrization, and version-specific code mapping.

Appendix A: Corrections and retained evidence boundaries

1. **Connection and derivative notation.** The source notes used the same state notation for Bloch functions and columns of U. Equations (16)–(19) distinguish them. The ordinary derivative $\partial_d X$ was also confused with the operator/comma derivative $X_{;d}$; Sec. II D repairs that distinction. The right-block gauge transformation has left index n, not the unrelated index m.

2. **Curvature and Hermiticity.** The anti-Hermitian relation is $(D^a_{jk})^* = -D^a_{kj}$. The z-curvature contraction uses $xy-yx$, not $xy-yz$. The internal trace curvature has the positive sign in Eq. (69). The $+2\,\mathrm{Im}\,Q$ convention in `QGD.tex` is replaced consistently by Eq. (9).

3. **BCP factors and denominator derivatives.** The metric symmetrization and the symmetric energy-denominator construction each contribute a factor of $1/2$. An expanded derivative in `Wannier.tex` lost one of these factors and repeated terms instead of exchanging all a, b indices. Equations (101)–(103) retain the two operations explicitly. The Sylvester identities supplement the single-band quotient rule without assuming diagonal intrablock velocity matrices. The archived `DerBCP` omits the terms in Eq. (96) by combining a multiband numerator derivative with diagonal energy-gradient differences. Section IV D gives the resulting BCP correction and explains the inverse-Sylvester derivative used in the corrected `calculators_collect` implementation.

4. **Incomplete QGQ expansion.** `Wannier.tex` leaves its part III empty, and `QGD.tex` repeats the part-I label on a different group of terms. These partial expressions are not reproduced as a complete result. Equation (112) records the full quotient identity, including the $+6$ denominator term and the band-energy Hessian difference.

5. **Orbital terms.** The energy-weighted Wannier matrix uses derivatives a, b, not b, b. Orbital-moment, magnetization, and interband magnetic-kernel symbols are separated. The original quadrupole expression mixed an outer $e/(2\hbar)$ with a magnetic matrix already containing that factor. Section VII B uses a single explicit reduced-unit convention.

6. **Exploratory derivations.** The unfinished ME1 projector manipulations, partial magnetization derivatives, and terminal Green-function identities are retained in the original notes. They are not needed for the reusable formula chain in this handbook. In particular, differentiation of a projected product must include all projector derivatives; differentiating only its explicit Hamiltonian is incomplete.

Appendix B: Consistency checks for later calculator validation

Useful algebraic checks include

$$\begin{aligned} Q^{ab}_n &= (Q^{ba}_n)^*, \qquad G^{aa}_n \geq 0, \\ \widetilde{F}^{ab:d}_{mn} &= (\widetilde{F}^{ba:d}_{nm})^*, \qquad \widetilde{F}^{ab:de}_{mn} = (\widetilde{F}^{ba:de}_{nm})^*, \\ \mathcal{G}^{ab}_n &= \mathcal{G}^{ba}_n, \\ \partial_d\partial_c X_n &= \partial_c\partial_d X_n \quad \text{for a smooth gauge-invariant scalar } X_n. \end{aligned} \tag{B1}$$

Compare the QGD band sums, matrix products, and centered finite differences on a gapped multiband model. Test the BCP block denominator identities under independent unitary rotations in both blocks. For Brillouin-zone integrals, compare sea/surface kernels only after matching axes, signs, units, occupations, and integration normalization. These checks do not replace convergence tests on the eventual material model.

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[2] Y. Gao, S. A. Yang, and Q. Niu, Field Induced Positional Shift of Bloch Electrons and Its Dynamical Implications, Phys. Rev. Lett. **112**, 166601 (2014).

[3] Y. Gao and D. Xiao, Orbital magnetic quadrupole moment and nonlinear anomalous thermoelectric transport, Phys. Rev. B **98**, 060402(R) (2018).